

BIOL550: Two SRA Datasets (Selection & Validation Write-up)

Piter Z. Garcia Bautista

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1 Chosen Datasets

The group chose three candidate RNA-seq projects and most members leaned toward two of them (**PRJNA1017789** and **PRJNA1277581**). For my individual submission, I replace the second pick with **PRJNA717662**, since it better matches my research interests. The two datasets I am proposing are: **PRJNA1017789** (mouse; [BioProject](#)) and **PRJNA717662** (human; [BioProject](#)).

2 Validation process (manual and reproducible)

I used SRA Run Selector RunInfo CSVs to validate: RNA-seq, ≥ 20 runs (used as the “samples” unit for initial screening), paired-end (`LibraryLayout=PAIRED`), depth $\geq 40,000,000$ **spots** per run, and read length consistent with the course requirement (`avgLength` near ~ 300 for paired-end $\sim 2 \times 150$). Note: for paired-end runs, **spots** is commonly a proxy for read pairs (often discussed as “fragments”), and `avgLength` is the average *spot length* (roughly the sum of both reads), not the library insert size.

3 Summary Table (final two datasets)

BioProject	Organism	Runs	avgLength (min–max; mean)	spots (min–max; mean)
PRJNA1017789	<i>Mus musculus</i>	26	302–302; 302.00	40,136,901–65,409,634; 46,078,299
PRJNA717662	<i>Homo sapiens</i>	152	287–300; 295.15	43,836,864–108,271,493; 67,329,439

Table 1: Run-level stats from local RunInfo CSVs. Depth is reported as **spots** (a proxy for read pairs/fragments).

4 Dataset Details (brief)

PRJNA1017789 (mouse) — axon regeneration after injury

Project accession: PRJNA1017789.

Brief description: RNA-seq study in a mouse injury/regeneration context, investigating how the aryl hydrocarbon receptor relates to axon regeneration in DRG neurons.

Sample count: 26 runs (GEO **GSM*** accessions in RunInfo); per-condition counts come from GEO sample metadata.

Case/control: injury vs control (and/or perturbations), as defined in GEO sample metadata.

Strengths (pros)

- Clear mechanistic biological question (injury response/regeneration) with a focused design.
- Uniform read-length proxy (`avgLength` = 302 for all runs) and all runs exceed the 40M-spot depth threshold.
- Moderate run count (26) is manageable for course timelines while still meeting the ≥ 20 sample requirement.

Weaknesses (cons)

- Model organism (mouse); translation to human clinical outcomes may be indirect.
- Case/control labels require pulling sample metadata from GEO (RunInfo alone is not enough).

PRJNA717662 (human) — long-term blood transcriptome after SARS-CoV-2

Project accession: PRJNA717662.

Brief description: Blood RNA-seq from healthy controls and recovered COVID-19 patients at 12, 16, and 24 weeks post-infection; the project reports long-lasting transcriptomic signatures linked to severity.

Case/control: healthy controls vs recovered COVID-19; additional grouping by timepoint (12/16/24 weeks) and disease severity (per GEO metadata).

Condition counts (from RunInfo Subject_ID): 14 healthy-control runs (14 subjects) and 138 recovered-COVID runs (65 subjects; repeated measures across timepoints, with some missingness).

Strengths (pros)

- Strong healthcare relevance (human cohort; post-infection immune signatures).
- Large sample size (152 runs) supports robust statistics and subgroup analyses.
- All runs meet the 40M-spot depth criterion; depth is high on average (mean $\approx 67,329,439$ spots).

Weaknesses (cons)

- `avgLength` varies (287–300); suggests variations in read length or trimming; pipeline should handle this consistently.
- Clinical covariates and subgroup definitions come from GEO metadata and must be curated carefully.

Overall, both datasets pass the screening criteria. PRJNA717662 is my stronger fit (human + clinically focused); PRJNA1017789 is a smaller, well-scoped mechanistic dataset.